



Title: RitmoForceCurve Sensor Setup & Calibration Guide

Introduction:

The RitmoForceCurve™ Sensor System measures real-time oar force, motion, and stroke dynamics to deliver precise performance feedback for rowers. Designed for durability and accuracy, each Bluetooth-enabled sensor captures high-resolution force curves and stroke timing, syncing seamlessly with the RitmoForceCurve mobile app. This guide provides setup, calibration, and troubleshooting instructions to ensure optimal sensor alignment and reliable on-water data collection.

1. Included Components (per sensor):

- RitmoForceCurve Sensor



Figure 1

- Silicone Band

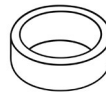


Figure 2

- Mounting Bracket



Figure 3

- Rubber Underpad

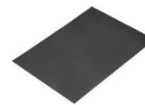


Figure 4

- Silicone Strap



Figure 5

Assembly Instructions:

1. **Seal the sensor:** Use the small silicone band to protect the USB-C port and sensor seam from water. Wrap it snugly around the sensor.

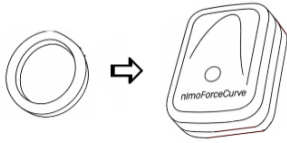


Figure 6

2. **Mount the sensor:** Insert the banded sensor into the plastic bracket. Ensure the sensor button faces away from the curved (underside) of the bracket. Double-check that the band remains aligned over the USB-C port and seam.

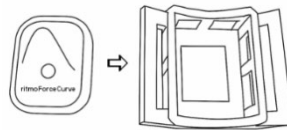


Figure 7

3. **Pad the oar:** Place the rubber pad between the bracket and the oar shaft. This provides grip and protects the oar surface.

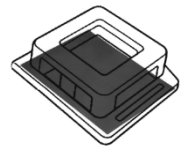


Figure 8

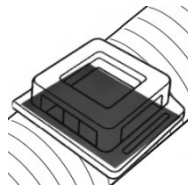


Figure 10

4. **Secure with Velcro/Silicone:** Use the silicone strap to fasten the bracket tightly to the oar. Use the best strap hole for the size of the oar (sweep or scull). Make sure that the brackets are not slipping on the oar shaft.



Figure 11

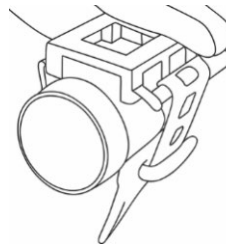


Figure 12

Sensor Orientation:

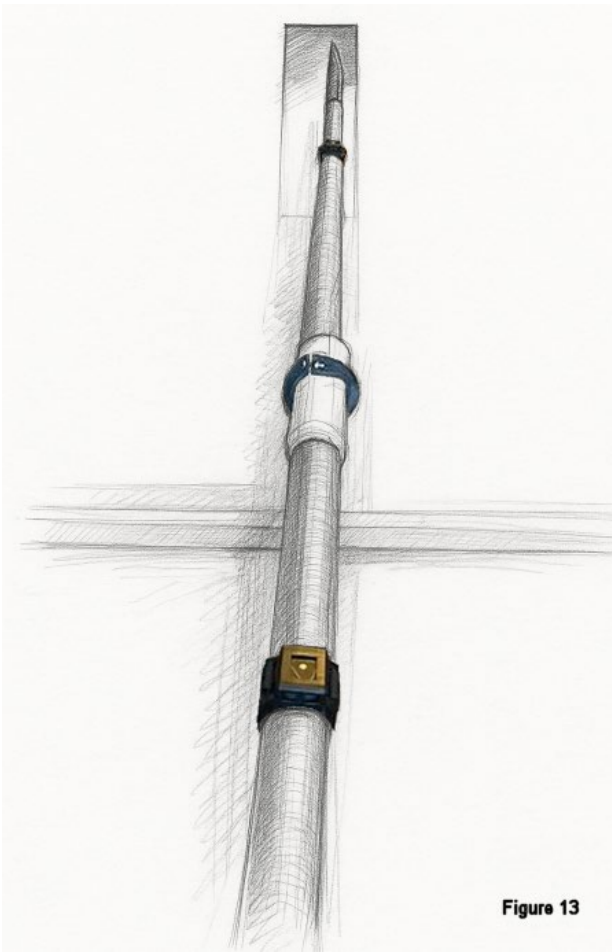


Figure 13



Figure 14

- USB-C port and “ritmoForceCurve” text need to face the blade.
- During the drive, the sensor button should be horizontal, during recovery, vertical. Mount the sensor this way before launch.
- Recommended sensor positions:
 - Sensor 1: Between the handle and the button.
 - Sensor 2: Roughly two-thirds of the way from the button to the blade.

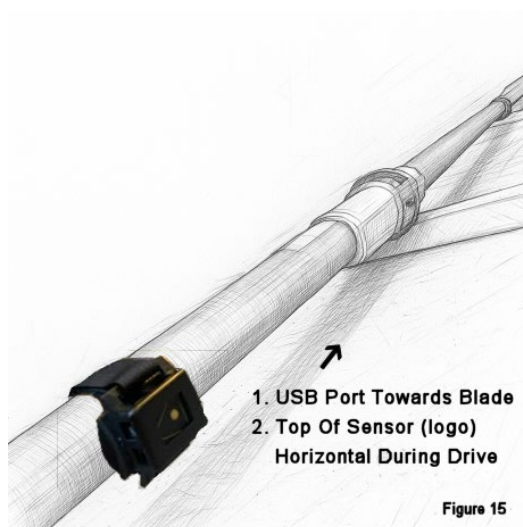


Figure 15

Connecting & Calibration

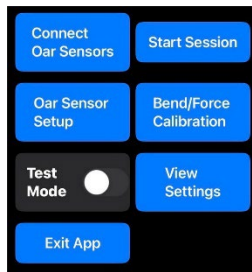


Figure 16

1. Connecting the Sensors:

- Turn on both sensors before placing the oars in the oarlocks or pushing off from the dock.
- Open the app → Main Menu → Connect Oar Sensors.

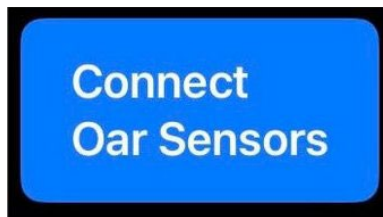


Figure 17

- Tap Turn on scanning for surrounding devices.

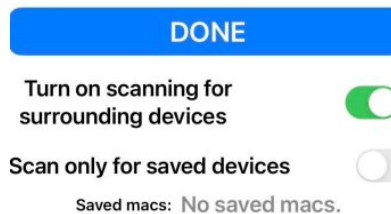


Figure 18

- If pairing for the first time, disable Scan only for saved devices.
- Select both sensors (they'll appear by MAC address), then tap DONE.

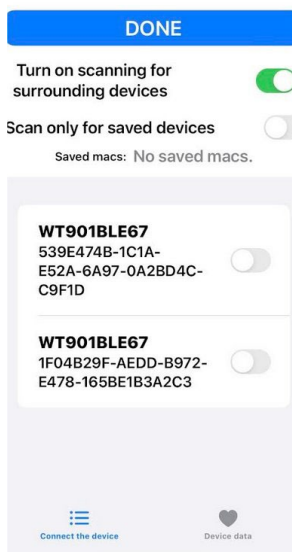


Figure 19

Oar Sensor Setup

- Choose port or starboard.
- To check Sensor alignment, place oar blade face up on level floor with something between the back of the sleeve and the floor that keeps the back of the sleeve parallel to floor.
- Check to make sure angles for both sensors are within -5 to 5 deg. Adjust if necessary. The app auto calibrates while you are rowing but needs them to be reasonably close.

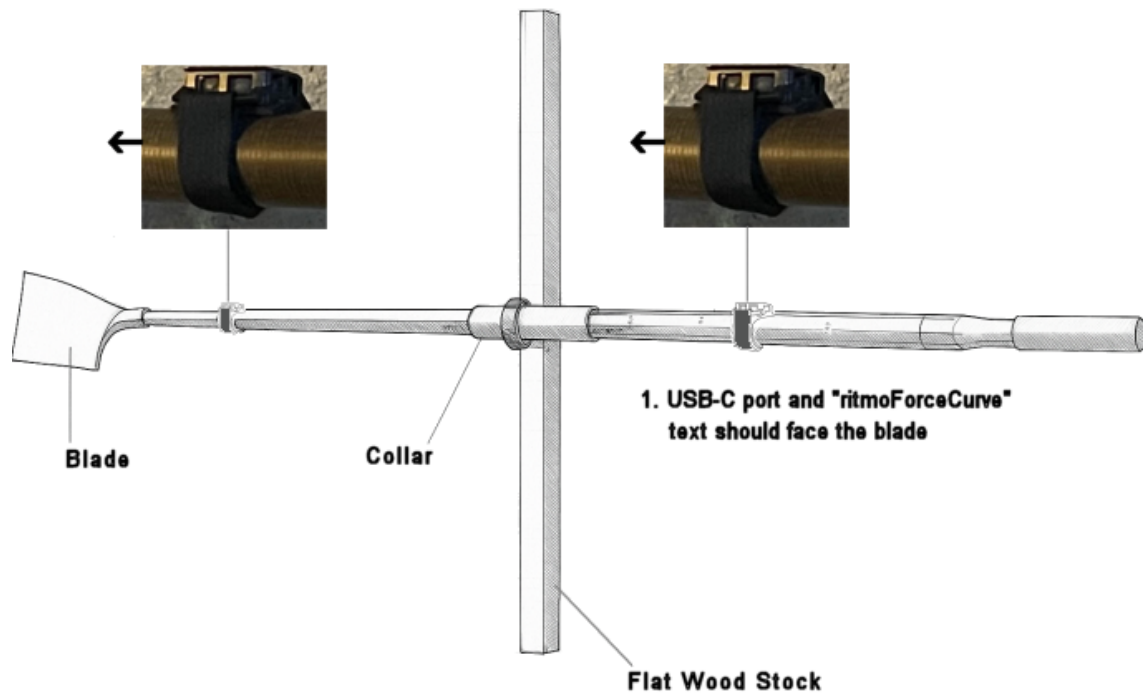


Figure 20

Bend Force Calibration:

- Scales Watts and ergSplit. Does not affect max bend, force curve shape, stroke length, slip, or wash.
- Bend readings vary based on oar stiffness and sensor placement.
- Manually adjust bend force value until ergSplit matches expected output



Figure 21

Troubleshooting Table:

Issue	Possible Cause	Solution
Sensor not detected	Bluetooth off	Turn on sensor / enable Bluetooth
Data spikes	Loose bracket	Check Velcro strap / reposition pad
Bend reading inaccurate	Misaligned sensor	Adjust orientation / recalibrate